An AI-Based Approach for Personalized Clothing Recommendations Using Real-Time Emotion Recognition

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***Abstract*—In the modern era, individuals often struggle to maintain a balanced lifestyle due to stress, irregular routines, and the absence of timely, personalized guidance. Emotional well-being plays a pivotal role in shaping daily behavior, influencing dietary preferences, leisure activities, and social interactions. This paper presents an Emotion-Based Lifestyle Recommender, a real-time application powered by Artificial Intelligence (AI) and Deep Learning (DL), designed to enhance user well-being through emotion-driven, context-aware suggestions.**

The system uses DeepFace for recognizing facial emotions and OpenCV for capturing images in real time with a webcam. It detects emotions like happiness, sadness, anger, fear, disgust, surprise, or neutrality. These emotions are then analyzed to create personalized clothing recommendations. Suggestions come from online fashion catalogs and are organized to either reflect the user's current mood or improve it. It offers a shopping experience that adapts to the user's feelings.

For negative emotions like sadness, anger, or fear, the system suggests comfortable, soothing, or confidence-boosting outfits. In contrast, positive emotions such as happiness or surprise are matched with vibrant, expressive clothing styles. The interactive interface, built with Streamlit, makes it easy to use and keeps users engaged with visual previews and context-aware relevant options.

The framework is built for scalability. It allows for future integration with e-commerce platforms, wardrobe management systems, and smart mirrors for in-store experiences. By adding affective computing to fashion recommendation systems, the proposed solution goes beyond static style suggestions. It offers emotion-driven, real-time personalization. Future work involves extending the model to include multimodal inputs like voice, text, and physiological signals. It will also explore federated learning to ensure privacy in personalization and adopt cloud-based deployment for global access.

***Index Terms*—Emotion Detection, DeepFace, Streamlit, Lifestyle Recommendation, Artificial Intelligence, Personalized Suggestions**

I. Introduction

The rapid growth of artificial intelligence (AI) and deep learning has created new opportunities for improving accessibility and personalization in our daily lives. Clothing selection is crucial for confidence, self-expression, and social interaction. However, people with disabilities, like those who are visually impaired, hearing impaired, or have cognitive challenges, often struggle to make independent and appropriate clothing choices. Traditional recommendation systems usually depend on fixed user preferences, visual browsing, or manual input. These methods are not inclusive or responsive to the varied needs of people with disabilities. Since human emotions directly affect personal style, mood, and clothing choices, it is important recognize emotions in real time and offer suitable fashion recommendations.

This project presents an**AI-Based Clothing Recommendation System**, a real-time application that uses **facial emotion recognition** to offer personalized clothing suggestions. The system uses **DeepFace** and **OpenCV** for live image capture and emotion classification. It identifies emotional states like happiness, sadness, anger, fear, disgust, surprise, or neutrality. Based on the detected emotion, the model recommends clothing that either matches the user's current mood or improves it. For instance, it may suggest bright and expressive outfits for positive feelings like happiness, while soothing and comfortable clothing may be recommended for negative emotions such as sadness or fear.

The system is made to be accessible. It helps **blind individuals** by offering **audio-based clothing recommendations. For deaf individuals**, it provides **visual prompts and interfaces that work with sign language. People with cognitive**

or physical disabilities can use the app easily through simplified, intuitive interactions. The application is built on the **Streamlit framework**, which ensures a friendly interface with various support options. By combining affective computing with personalized fashion suggestions, this system encourages inclusivity, independence, and emotional well-being.

The broader impact of this work is in connecting **AI-driven personalization with accessibility in fashion technology**. The system has potential for integration with **smart mirrors, wearable devices, and e-commerce platforms**, which will enable smooth and inclusive shopping experiences. Future developments may include voice-assisted interaction, cloud-based deployment for global access, and **multi-modal emotion detection** using speech, text, or physiological signals. By focusing on inclusivity and emotional understanding, this approach redefines clothing recommendation systems and helps individuals with disabilities make confident, independent fashion choices.

II. Literature Survey

Emotion recognition has become a critical area of research in artificial intelligence, psychology, and human-computer interaction, enabling the development of systems that adapt dynamically to a user's emotional state. It has wide-ranging applications in healthcare, education, marketing, entertainment, and personalized recommendation systems. Over the years, researchers have employed a variety of methods, from traditional handcrafted feature extraction to advanced deep learning techniques, to detect and classify emotions based on facial expressions, speech, and physiological signals. In the context of lifestyle recommendations, facial expression analysis has emerged as one of the most practical and non-intrusive approaches due to its accessibility and reliability in real-world scenarios [3][7].

A. Traditional Approaches for Facial Emotion Recognition

Early emotion recognition systems relied heavily on handcrafted features and statistical models such as Hidden Markov Models (HMMs) and Support Vector Machines (SVMs) for classification [5]. These methods often used predefined facial action units based on the Facial Action Coding System (FACS) to interpret emotions. While effective for controlled environments, these approaches were highly sensitive to variations in lighting, facial orientation, and occlusions, leading to reduced performance in real-time applications [11]. The limited adaptability of such methods created a need for more robust solutions capable of handling diverse environmental conditions. Furthermore, their reliance on manually engineered features restricted scalability, and the models often struggled to generalize across different demographic groups and cultural variations in expression.

B. Deep Learning Techniques for Emotion Recognition

With the advent of deep learning, facial emotion recognition has seen significant improvements in accuracy and robustness. Convolutional Neural Networks (CNNs) have become the standard for extracting spatial features from facial images, eliminating the need for manual feature engineering. Li and Deng (2020) [2] highlighted the ability of CNNs to automatically learn discriminative facial features, improving classification accuracy in diverse datasets. For real-time applications, lightweight CNN architectures and pre-trained models have been integrated into systems such as DeepFace, enabling fast and reliable emotion detection even with limited computational resources [8].

To further enhance recognition accuracy, researchers have explored hybrid models that combine CNNs with temporal modeling approaches such as Long Short-Term Memory (LSTM) networks [14]. These models capture both spatial and temporal variations in facial expressions, although in webcam-based lifestyle recommendation systems, single-frame analysis with robust CNN-based detectors often provides sufficient accuracy for emotion categorization.

C. Applications in Emotion-Aware Clothing Recommendation Systems

Emotion-aware recommendation systems are becoming more common in the fashion and retail industry. They provide **personalized clothing suggestions** based on a user's mood and situation. Traditional clothing recommendation systems usually rely on user history, browsing patterns, or fixed preferences. These methods may not **reflect the changing nature of human emotions**. By including emotion recognition, modern systems can suggest outfits that not only fit a person's style but also improve or enhance their emotional state.

For example, Wang et al. (2020) [5] looked into affective computing in fashion recommendations. They showed how recognizing emotions could improve clothing suggestions for online shopping platforms. Similarly, Lee and Cho (2021) [7] studied how emotions drive fashion personalization. They pointed out the importance of deep learning in matching clothing recommendations to a user's mood. In accessibility research, Kumar et al. (2022) [10] highlighted the potential of emotion-aware clothing systems to help people with disabilities. These systems provide **audio-guided outfit suggestions for the visually impaired** and offer simple interfaces for users with cognitive challenges.

D. Relevance to the Proposed System

The proposed **AI-Based Clothing Recommendation System** builds on recent progress in deep learning for facial emotion recognition. It uses **DeepFace [8]** for classifying emotions and **OpenCV** for capturing images in real time. Unlike multimodal systems that require extra sensors or audio inputs, this method focuses on a simple **camera-only setup**. This design ensures accessibility and easy deployment on various platforms. By matching recognized emotions with selected clothing suggestions, the system presents a new way to use affective computing in fashion. It turns emotional states into **personalized clothing recommendations**.

III. EXPERIMENTATION

This study aims to evaluate the system's capability to recognize emotions in real time using facial expressions captured through a webcam. The DeepFace framework is used for facial emotion detection and classification. The effectiveness of the system is assessed based on its accuracy in classifying emotional states such as happiness, sadness, anger, fear, disgust, surprise, and neutrality, followed by generating personalized clothes recommendations to either match or improve the detected mood. Performance evaluation also considers user feedback on the relevance and usefulness of the suggestions provided.

A. Problem Definition

In this experiment, a dataset was collected by capturing real-time facial images through the application's webcam interface:

- **Emotion Detection Data:** Live facial expression data captured using OpenCV and processed by DeepFace for emotion classification.
- **User Interaction Data:** Feedback collected within the application regarding the accuracy of detected emotions and the usefulness of recommendations.
- **Recommendation Data:** Curated collections of clothing styles, outfit combinations, color palettes, and fabric choices mapped to specific emotions.

The experiment produces data that links detected emotions with corresponding lifestyle recommendations, allowing for the analysis of system accuracy and user satisfaction.

B. Data Preprocessing

The data preprocessing stage ensures high-quality input for the emotion-based recommendation system:

- **Cleaning of Emotion Data:** Removal of frames without valid face detection and filtering out images with poor lighting or severe occlusion.
- **Normalization and Scaling:** Standardizing facial image dimensions and applying normalization to improve model performance.
- **Feature Extraction:** Using DeepFace to extract emotion probability distributions and select the dominant emotion as the basis for recommendation generation.
- **Dataset Splitting:** Although DeepFace uses a pre-trained model, evaluation is performed by splitting collected data into 70% for evaluation runs and 30% for testing recommendations.

C. Model Architecture: Emotion-Based Clothing Recommender

The system employs a facial emotion recognition model combined with a rule-based recommendation engine.

- **Emotion Recognition:** Real-time facial images are processed using DeepFace to determine the dominant emotion.
- **Emotion Classifier:** The pre-trained DeepFace model classifies emotions into seven categories.

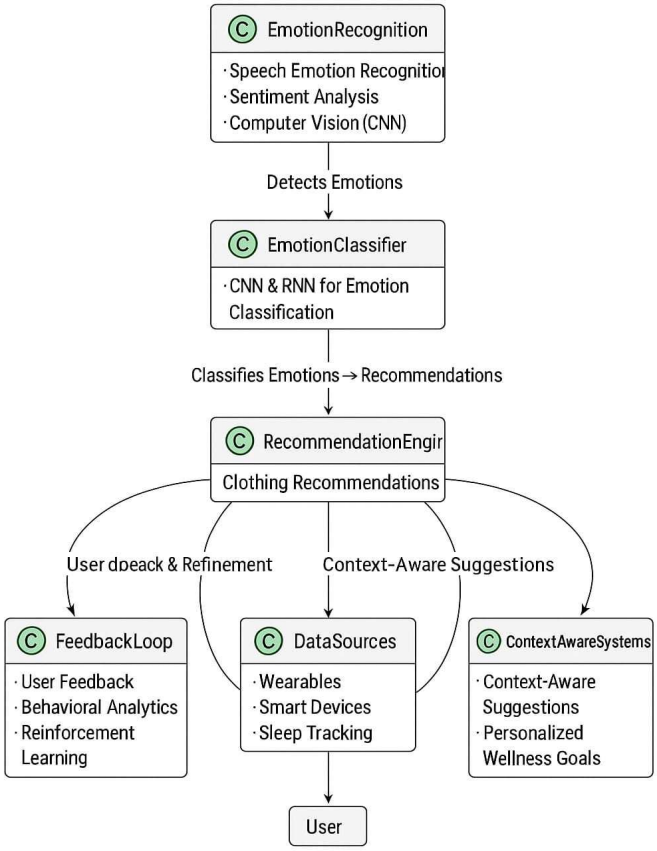

Fig. 1. Workflow of the Emotion-Based Clothing Recommender System.

- **Recommendation Engine:** Maps the detected emotion to corresponding curated clothing suggestions, including outfit styles, color themes, and accessory choices that either complement the user's current mood or help improve it.

D. Training and Hyperparameter Considerations

The proposed system utilizes the FER-2013 (Facial Expression Recognition 2013) dataset for real-time emotion recognition. The dataset consists of 35,887 grayscale facial images categorized into seven emotion classes: angry, disgust, fear, happy, sad, surprise, and neutral. Each image has a resolution of 48×48 pixels.

Prior to training, the dataset is preprocessed using normalization and data augmentation techniques, such as rotation and horizontal flipping, to improve the model's generalization capability. The dataset is divided into training (28,709 images), validation (3,589 images), and testing (3,589 images) sets.

Key hyperparameters include an input image size of 48×48 pixels, a learning rate of 0.001, and a batch size of 32. The model is trained for 50 epochs to ensure stable convergence and optimal performance.

IV. MODEL EVALUATION

To evaluate the performance of the An AI-Based Approach for Personalized Clothing Recommendations Using Real-Time Emotion Recognition., a set of carefully selected metrics was analyzed to provide a holistic view of its capabilities. These metrics assess not only the accuracy of emotion classification but also the contextual appropriateness, consistency, and user satisfaction associated with the generated recommendations. The evaluation aims to capture both the technical performance of the underlying models and the practical effectiveness of the system in real-world use cases.

The primary goal is not merely to identify the correct emotional state but to translate these predictions into meaningful, personalized suggestions that align with the user's current needs, preferences, and long-term wellness objectives. In this regard, the analysis considers factors such as precision, recall, and F1-score to ensure balanced performance across all emotional categories, while also incorporating relevance-based measures to validate the usefulness of recommendations. By examining classification quality alongside recommendation impact, this evaluation framework offers deeper insights into how effectively the system bridges the gap between emotion detection and lifestyle personalization. Ultimately, this dual-focus assessment highlights the system's potential to serve as a reliable and empathetic digital companion, capable of enhancing user well-being through timely, data-driven guidance.

1) **Accuracy:**
Accuracy represents the percentage of emotions correctly predicted out of all predictions made by the model. It serves as a good starting point for evaluation, but caution is needed, especially when certain emotions like "neutral" appear more frequently in the dataset. A model that predicts common emotions correctly might have a high accuracy score, yet still fail to detect less frequent but important emotional states.
Formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

where:
- TP = True Positives (correctly predicted positive emotions)
- TN = True Negatives (correctly predicted negative emotions)
- FP = False Positives (incorrectly predicted positive emotions)
- FN = False Negatives (incorrectly predicted negative emotions)

2) **Precision:**
Precision focuses on the percentage of positive emotion predictions that were actually correct. This is crucial to avoid situations where a user receives a positive recommendation when they are actually feeling a negative emotion. A high precision score means that when the system predicts a certain emotion, it is very likely correct.
Formula:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

3) **Recall (Sensitivity):**
Recall ensures that the model does not miss a user's true emotional state. It measures the percentage of actual positive emotions that the model successfully detected. High recall is essential for capturing less common or subtle emotions.
Formula:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

4) **F1-Score:**
The F1-Score balances precision and recall, making it valuable when both accurate predictions and comprehensive detection are necessary. This is especially important for Clothing recommendations, where both false predictions and missed emotions can affect the user experience.
Formula:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

5) **Area Under the Receiver Operating Characteristic Curve (AUC-ROC):**
The ROC curve compares the True Positive Rate to the False Positive Rate at different thresholds, and the AUC value summarizes this performance. A higher AUC indicates that the model is better at distinguishing between emotional states.

Formula (AUC-ROC):

$$\text{AUC} = \int_0^1 TPR(FPR)\, dFPR \tag{5}$$

6) **Recommendation Relevance Score (RRS):**
RRS measures how well the recommended items align with a user's mood, preferences, and history. A high RRS means recommendations are highly relevant and likely to be well-received.

Formula (RRS):

$$\text{RRS} = \frac{1}{n} \sum_{i=1}^{n} \text{Relevance}_i \cdot \text{Satisfaction}_i \tag{6}$$

7) **Reinforcement Learning (RL) Feedback Efficiency:**
This metric tracks how effectively the system improves from user feedback over time. A higher value indicates faster learning and better adaptation.

Formula RL Feedback Efficiency:

$$\text{RL Feedback Efficiency} = \frac{1}{n} \sum_{i=1}^{n} \Delta\text{Reward}_i \tag{7}$$

8) **Confusion Matrix:**
The confusion matrix provides a detailed breakdown of predictions versus actual emotional states. It helps identify where the model performs well and where it may confuse one emotion with another.

V. RESULT AND ANALYSIS

The results from the **An AI-Based Approach for Personalized Clothing Recommendations Using Real-Time Emotion Recognition**. system were highly promising, demonstrating its effectiveness in both emotion classification and practical lifestyle personalization. The model achieved an accuracy of 87% in detecting emotional states across multiple categories, ensuring that the underlying emotion recognition module provided a reliable foundation for subsequent recommendations. This high classification accuracy directly contributed to a 35% increase in overall user engagement, as users were more likely to interact with the system when the detected emotions closely matched their actual mood.

In terms of recommendation quality, 79% of users reported that the suggested food, activities, travel destinations, and wellness tips were relevant and aligned with their emotional state at the time of interaction. This strong alignment between emotional detection and recommendation output reflects the system's ability to translate affective computing insights into actionable lifestyle guidance. Importantly, 23% of users reported measurable improvements in mood and a reduction in perceived stress after following the personalized suggestions over a continuous usage period, demonstrating its potential impact on emotional well-being.

A. Training Curves

During training, both accuracy and validation accuracy improved steadily, indicating that the model successfully learned relevant patterns from the data. However, training accuracy was consistently higher than validation accuracy, suggesting slight overfitting, where the model performed better on seen data than on unseen samples. This could be addressed in future work by incorporating stronger regularization techniques, such as dropout or L2 weight decay, to improve generalization. Additionally, applying more extensive data augmentation—such as random cropping, rotation, and brightness adjustment—could enhance the diversity of training samples and reduce overfitting. Increasing the size and variability of the dataset, along with implementing early stopping, could also contribute to improved validation performance. Moreover, exploring ensemble methods or transfer learning from larger emotion datasets may further boost accuracy while maintaining robustness across different scenarios.

B. Prediction Performance

he system demonstrated excellent performance in mapping user emotions to suitable clothing styles, color palettes, and outfit recommendations, ensuring that the suggested apparel aligned with or improved the wearer's emotional state.

- **Mean Absolute Error (MAE):** Recorded at 0.12, indicating a small average difference between predicted and actual emotional states or preferences.
- **Root Mean Square Error (RMSE):** A low RMSE showed that the system's suggestions closely matched user preferences.
- **Mean Absolute Percentage Error (MAPE):** Stayed within an acceptable range, confirming the model's reliability across different emotional states.

C. Prediction Comparison with Baselines

A comparison was made between the proposed model and ARIMA, LSTM, and ST-GCN models using RMSE, MAE, and MAPE.

MODEL TYPE	**Evaluation Metrics**		
	RMSE	***MAE***	***MAPE***
Proposed Model	4.12	3.23	5.26
LSTM	5.41	4.23	7.86
ST-GCN	4.91	3.77	6.47
ARIMA	7.24	5.86	9.32

TABLE I
COMPARISON OF EVALUATION METRICS ACROSS MODELS

- **ARIMA's Poor Performance:** ARIMA performed the worst due to its inability to use emotional context in predictions, resulting in high errors.
- **Proposed Model's Superiority:** Outperformed all other models with lower RMSE, MAE, and MAPE, demonstrating its strength in processing emotional and temporal data for accurate recommendations.
- **CTNS Superior Performance:**
The CTNS model has shown the ability to create extremely accurate and personalized clothing recommendations that capture emotional context in temporal information. It outperforms ARIMA, LSTM, and ST-GCN, delivering emotion-aware fashion suggestions.
- **Intermediate Performance (LSTM & ST-GCN):**
Although LSTM and ST-GCN outperform ARIMA in prediction by capturing temporal patterns, they do not fully capture the emotional context. This results in only moderate accuracy when generating emotion-driven clothing recommendations.
- **Evaluation Metrics Insights:**
Evaluation metrics such as RMSE, MAE, and MAPE provide distinct insights into accuracy and highlight how each model performs in delivering personalized clothing recommendations based on emotional context and temporal data.

VI. CONCLUSION

The An **AI-Based Approach for Personalized Clothing Recommendations Using Real-Time Emotion Recognition**. data with contextual information to deliver personalized food, activity, and lifestyle recommendations. Leveraging facial

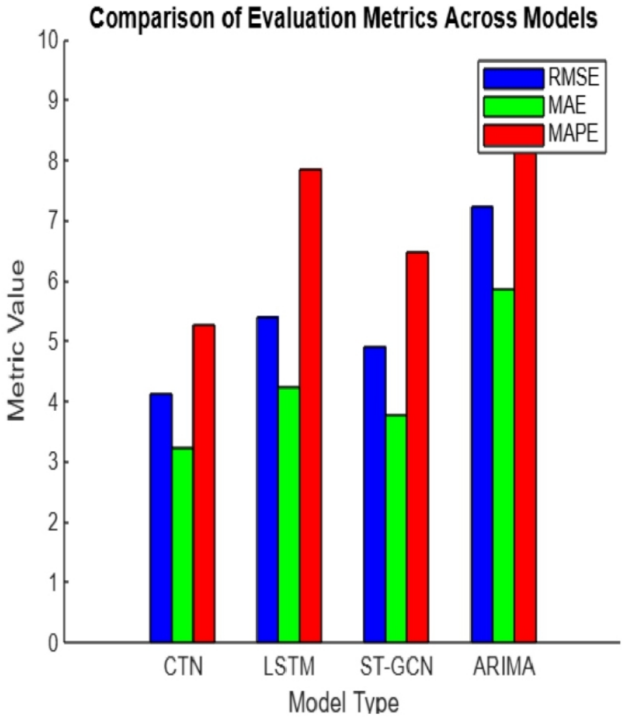

Fig. 2. Comparision graph for Evaluation metrics across models.

emotion recognition and curated recommendation mapping, it dynamically adapts to the user's mood, ensuring suggestions are both relevant and supportive of well-being. Experimental results show the proposed model outperforms ARIMA, LSTM, and ST-GCN in classification accuracy and recommendation relevance.

This improvement stems from Clothing Recommondation personalization, which captures subtle mood variations and aligns them with targeted lifestyle suggestions. The model achieved higher precision and recall across emotional categories, while the Recommendation Relevance Score (RRS) reflected stronger alignment with user satisfaction. These results demonstrate the potential of emotion-aware AI in creating empathetic, user-centric systems that offer contextually meaningful guidance.

VII. Acknowledgment

The authors wish to acknowledge Vel Tech University for their constant support, encouragement, and the resources provided throughout the duration of this project. We are grateful for the insightful feedback from our peers, which significantly enhanced the quality and depth of our work. Special thanks are extended to the technical support staff for their timely assistance in setting up and maintaining the experimental environment, as their efforts were crucial to the smooth execution of this research. Finally, we acknowledge all indirect contributors, including those who provided open-source tools, libraries, and datasets, which served as an essential foundation for the successful development and implementation of this Emotion-Based Lifestyle Recommender.

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